

MID TERM EXAMINATION
B. TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
 2nd Semester, April, 2024

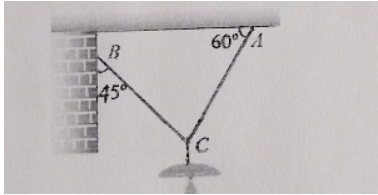
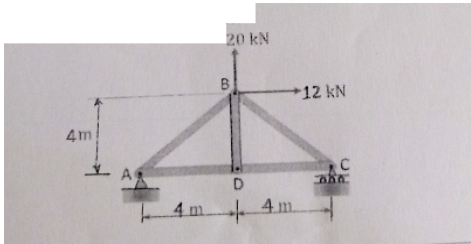
Paper Code: ES 114

Subject: Engineering Mechanics

Time: 1½ Hrs.

Max. Marks: 30

Note: Attempt Q.No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Questions	Marks	CO
1	Write short notes on the following: (a) What is the purpose of free body diagram? Explain with a suitable example. (b) Define centroid, center of mass and center of gravity. Under which conditions is the center of mass different from center of gravity? (c) What is a truss structure? What is the condition for a perfect truss? (d) State the laws of dry friction.	2.5 2.5 2.5 2.5	CO1 CO1 CO2 CO2
2	An electric light fixture weighing 15 N hangs from a point C, by two strings AC and BC. The string AC is inclined at 60° to the horizontal and BC at 45° to the horizontal as shown in figure. Determine the forces in the strings AC and BC. 	10	CO1
3	Using the method of joints, determine the axial force in AB, BD and BC members of the truss shown in the figure. 	10	CO2
4	Define angle of friction and angle of repose. Show that angle of friction is equal to angle of repose for a rough inclined plane.	10	CO2